



SOCIAL
IMPACT
CELL



ECOVATE

EnerGen

TEAM AIVELLE

TERRA NOVA

POWERING
TOMORROW

Can Garbage Become India's Fuel?

A WASTE-TO-FUEL BUSINESS CASE

A DATA-BACKED, POLICY-ALIGNED, IMPACT-DRIVEN

ONE ECOSYSTEM. TWO FUELS. UNLIMITED IMPACT

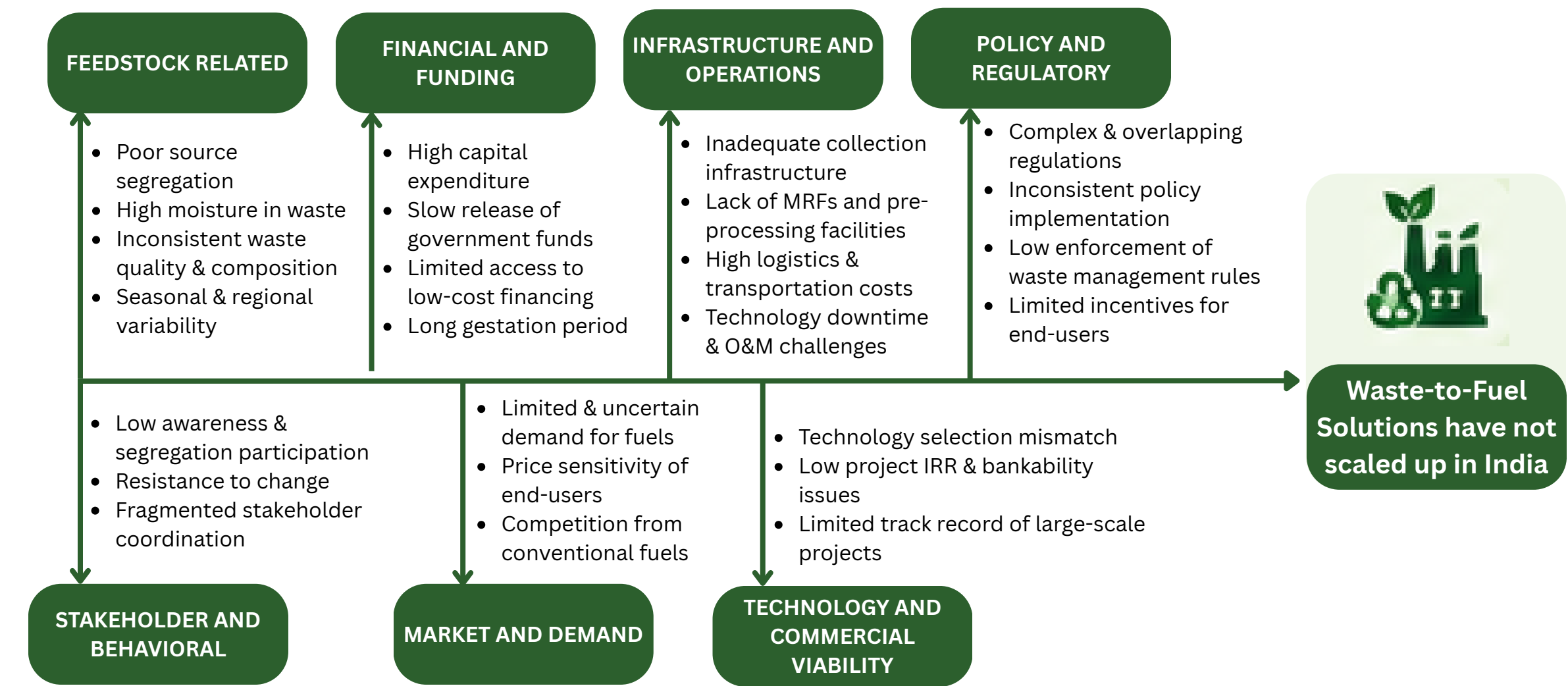
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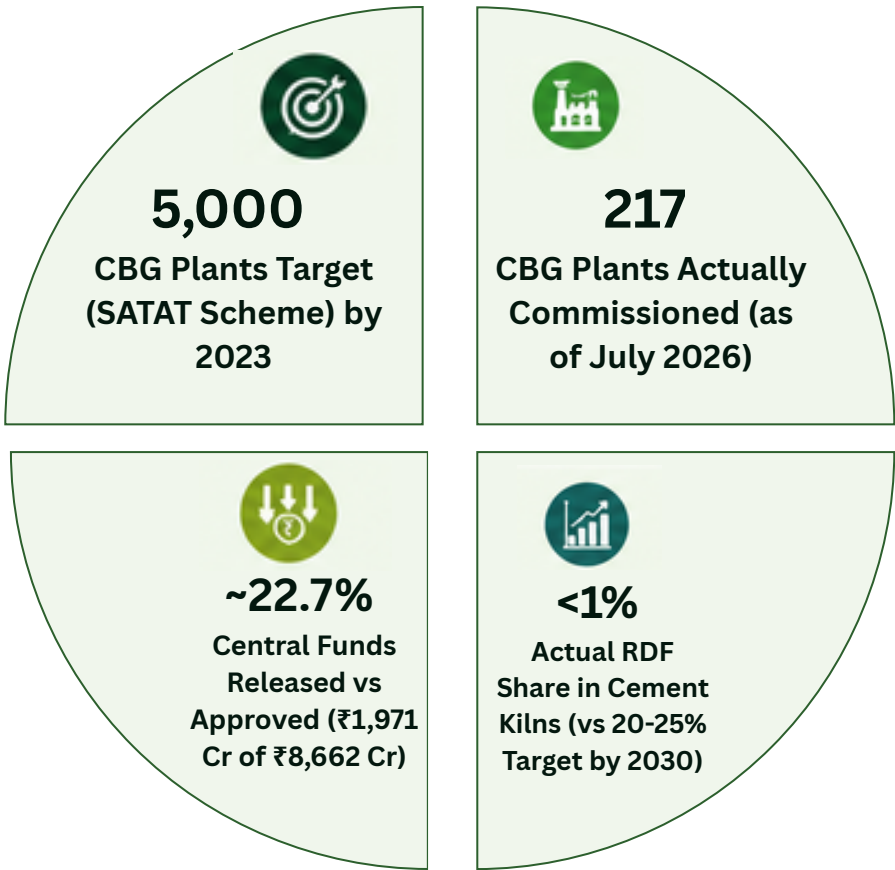


ROOT CAUSE ANALYSIS

Why Waste-to-Fuel Solutions Have not Scaled in India?



THE BIGGEST GAPS



The core issue is not the lack of policy or technology, but execution gaps across the value chain.

5-WHYS ANALYSIS



Every day, 31,449 tonnes of urban municipal waste remain untreated - equivalent to over 11 million tonnes annually, while industries continue to depend on fossil fuels. Bridging this gap presents one of India's largest untapped circular economy opportunities.

MARKET OPPORTUNITY

Strong Policy Push. Huge Demand. Timely Market.



161,157 TPD:

Urban MSW Generated
Every Day in India



~31,449 TPD:

Still Unprocessed Every
Day



₹23,549 Cr:

Total Project Value
Approved (SBM-U 2.0 –
SWM Component)



₹1,971 Cr:

Central Share Released
(Only 22.7% of Approved)



Significant waste volume, strong
government commitment and urgent need
for sustainable solutions create a massive
market opportunity.

INDUSTRY DEMAND IS ACCELERATING

CEMENT INDUSTRY: A READY MARKET FOR RDF



Cement sector is actively looking for alternative, cost-effective & sustainable
fuels.

- **SWM Rules 2026:** Mandatory RDF adoption for cement plants
- **RDF Replacement Mandate:** 5% baseline → 15% within 6 years
- **From 2031:** Plants within 400 km of RDF facility must meet ≥15% of fuel
needs via RDF
- **NITI Aayog 2026 Roadmap:** 20–25% Thermal Substitution Rate (TSR) by
2030

*Yet, actual RDF share in Indian cement kilns is <1%
Huge untapped demand waiting to be unlocked.*

CBG OPPORTUNITY IS MASSIVE

SATAT SCHEME TARGET vs REALITY



Target

5,000

CBG Plants by 2023

VS



Actual

217

CBG Plants Commissioned
as of July 2026

174

Units under
construction

425

Units yet to
begin
construction

Only ~4% of the target achieved. Enormous headroom for growth and first-mover advantage.

POLICY & FUNDING TAILWINDS



SBM-U 2.0

Strong focus on scientific
waste management and
resource recovery



MNRE Support

Attractive CFA for
CBG plants and
biogas-based projects



SWM Rules 2026

Mandatory segregation,
RDF usage & reduced
landfilling



Net Zero 2070

Commitment

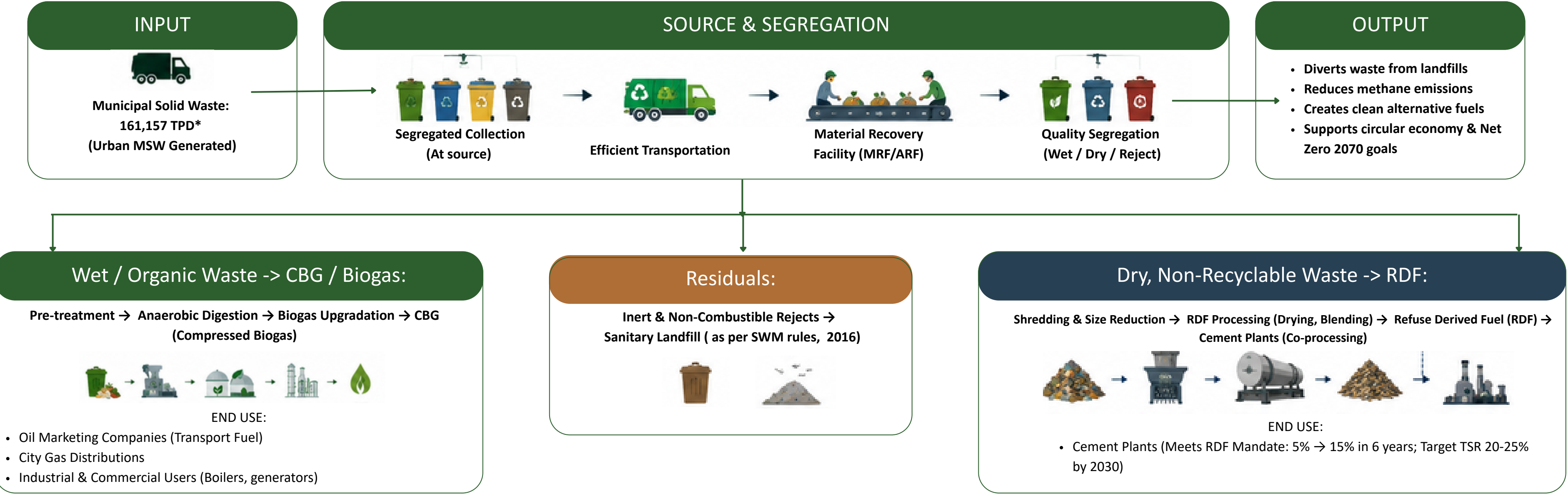
Driving shift towards circular
economy & alternative fuels

***Policy, Demand &
Capital are aligned like
never before - the time
to build is NOW!***

OUR SOLUTION

A Hybrid Waste-to-Fuel Ecosystem for a Cleaner, Greener Tomorrow

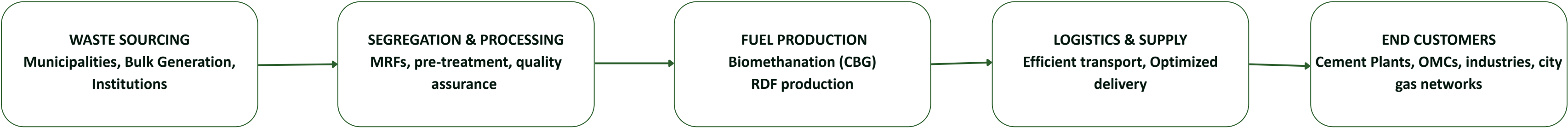
FROM WASTE TODAY, TO FUEL TOMORROW:
Maximizing value from every ton of waste through integrated & sustainable solutions



OUR BUSINESS MODEL

Multiple Revenue Streams. Strong Economics. Scalable Impact.

INTEGRATED WASTE-TO-FUEL VALUE CHAIN DRIVING SUSTAINABLE & PROFITABLE GROWTH



REVENUE STREAMS

TIPPING FEE INCOME: Paid by municipalities & large waste generators for scientific processing of waste.	CBG / BIOGAS SALES: Sale of Compressed Biogas (CBG) to Oil Marketing Companies (OMCs), city gas distribution networks, and industrial users.	RDF SALES: Supply of Refuse-Derived Fuel (RDF) to cement plants and other industries as a cost-effective, alternative fuel.	FOM / BY-PRODUCT SALES: Sale of by-products such as compost, recovered recyclables, and inert materials.
REVENUE DRIVER: waste volume (kg/day) * tipping fee(rs/kg)	REVENUE DRIVER: CBG output (kg/day) * Realization (rs/kg)	REVENUE DRIVER: RDF sales (kg/month) * price (rs/kg)	REVENUE DRIVER: FOM output (kg/month) * price (rs/kg)

COST STRUCTURE

 • FEEDSTOCK COLLECTION & LOGISTICS: Waste collection, transportation, handling & weighbridge
 • PROCESSING & OPERATIONS: Segregation, treatment, plant O&M, utilities & consumables
 • MANPOWER & ADMINISTRATION: Salaries, admin, compliance & overheads
 • MAINTENANCE & OTHER COSTS: Equipment maintenance, spares, insurance & misc.

BUSINESS MODEL HIGHLIGHTS

 MULTIPLE REVENUE STREAMS: Reduces dependence on a single source	 ASSET-LIGHT PARTNERSHIPS: Strategic partnerships with ULBs, cement plants & OMCs	 SCALABLE & REPLICABLE: Modular plants & hub-and-spoke model for rapid expansion
 POLICY BACKED: Aligned with SBM 2.0, MNRE support & SWM Rules 2026	 SUSTAINABLE & IMPACTFUL: Lower landfill, reduced emissions, circular economy enabler	 Long-Term Contracts: Offtake agreements ensure revenue visibility

KEY VALUE PROPOSITION

Converts waste management cost into energy value
Creates multiple revenue streams for strong financial resilience
Supports industries with reliable, cost-effective alternative fuels
Enables municipalities to achieve sustainable waste management goals
Drives measurable environmental & social impact

IMPLEMENTATION ROADMAP

Mission: Build a Scalable, Profitable & Sustainable Waste-to-Fuel Ecosystem Across India

PHASE	PILOT & FOUNDATION	INFRASTRUCTURE BUILD-UP	MARKET EXPANSION	SCALE & OPTIMIZATION	NATIONAL SCALE
	0-6 MONTHS TEST . LEARN . VALDATE	6-18 MONTHS STRENGTHEN . EXPAND . INTEGRATE	18-36 MONTHS SCALE . PARTNET . MONETIZE	36-60 MONTHS OPTIMIZE . STANDARDIZE . EXCEL	60+ MONTHS LEAD . IMPACT . TRANSFORM
KEY INITIATIVE	Establish pilot operations and validate technology, feedstock sourcing & partnerships. - Select 1–2 cities (100–300 TPD) - Secure land, statutory approvals - Setup pilot CBG & RDF units - Build waste collection partners - Sign anchor offtake MoUs	Build core infrastructure and integrate waste sourcing & processing systems. - Scale MRFs & pre-processing units - Expand waste collection network - Commission CBG plant - Establish RDF production line - Implement digital monitoring	Expand market reach and secure long-term off-take partnerships. - Sign long-term offtake contracts with cement plants & OMCs - Expand to 3–5 additional cities - Optimize logistics & distribution - Strengthen brand & visibility	Improve operational efficiency and achieve cost leadership across plants. - Standardize plant design & SOPs - Implement advanced process optimization - Improve yield & energy efficiency - Centralize procurement & O&M	Build a pan-India presence & become the leading integrated WtF enterprise. - Expand to 15+ cities / clusters - Explore export of technology & expertise - Invest in R&D & innovation - Drive policy advocacy & sector leadership
KEY OUTCOMES	Operational pilot plants, validated economics & strong stakeholder buy-in	Integrated operations with reliable supply chain & steady fuel production	Stronger market penetration, long-term revenue visibility & diversified customer base	Cost-efficient, high-yield operations with strong unit economics	National leadership in waste-to-fuel with sustainable impact at scale



ENABLERS FOR SUCCESS



STRATEGIC PARTNERSHIPS:
Municipalities, ULBs, cement plants, OMCs



TECHNOLOGY & INNOVATION:
Advanced processes, digital & automation



STRONG FINANCIAL STRUCTURE:
Blended finance, CFA & green funds



TALENT & CAPABILITIES:
Skilled teams across operations & tech



ESG & COMMUNITY ENGAGEMENT:
Responsible operations, shared value creation

KEY MILESTONES



PILOT LIVE:
100–300 TPD pilot operational



1ST CBG PLANT COMMISSIONED:
Steady biogas/CBG production



RDF SUPPLY CONTRACTS LIVE:
Long-term offtake secured



5+ CITIES OPERATIONAL:
Multi-city integrated operations



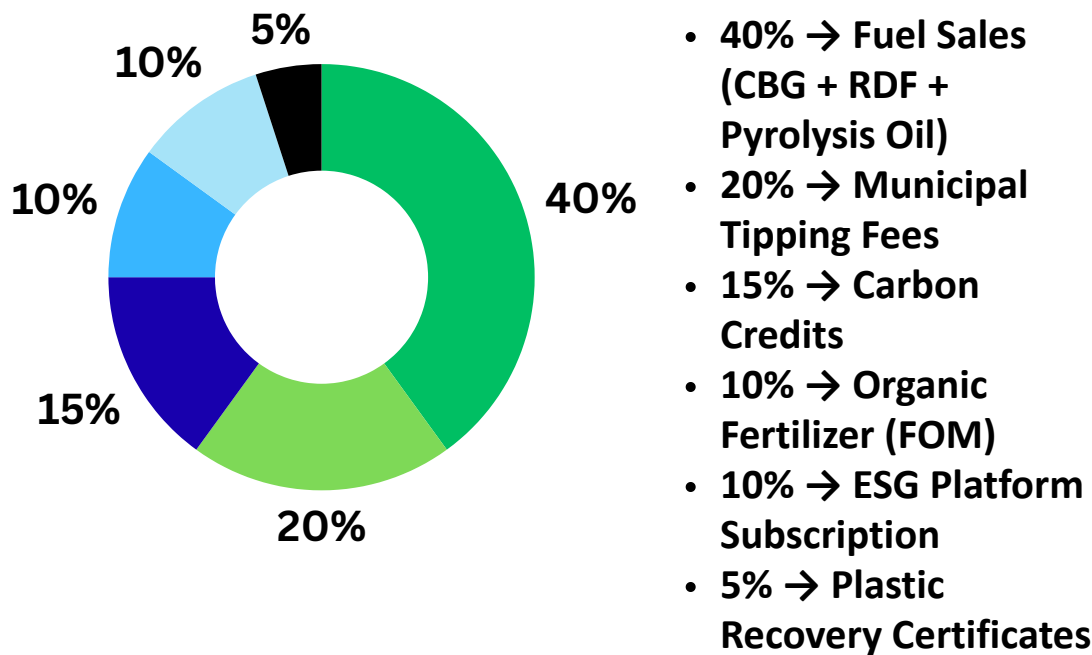
NATIONAL LEADER IN WtF:
Scalable, profitable & sustainable impact

From Pilot to Scale: Building India’s Leading Waste-to-Fuel Ecosystem

A phased, partnerships-driven approach to build a future-ready, financially strong, and environmentally meaningful enterprise.

FINANCIAL SNAPSHOT

REVENUE MODEL



Six independent revenue streams reduce dependence on any single customer or product.

GOVERNMENT SUPPORT & FINANCIAL ENABLERS

Policy / Scheme	Key Provision	Business Benefit to EnerGen
SATAT Scheme	Long-term Commercial	Guaranteed market demand
Revised CBG Pricing	₹1,478/mmBtu (~₹77.4/kg) procurement price	Predictable revenue and
Transportation Incentive	₹1.5–₹2.5/kg for longer delivery distances	Reduces logistics costs
Compression Support	₹8/kg compression charge	Improves operating margins
MNRE Capital Subsidy	Up to ₹10 Cr/project for eligible CBG projects	Lowers initial CAPEX
Priority Sector Lending (PSL)	Loans up to ₹50 Cr	Easier access to low-cost financing
CBG Blending Obligation	Mandatory blending for	Expands long-term demand for CBG
Excise Duty Exemption	Tax relief on CBG	Enhances profitability

ILLUSTRATIVE PILOT ECONOMICS

Metric	Value
Processing Capacity	100 TPD
Estimated CAPEX	₹35–43 Cr (Illustrative)
Government Subsidy	Up to ₹10 Cr
Target EBITDA Margin	>20%
Expected IRR	18–22%
Estimated Payback	4.5–5.5 Years

Illustrative estimates based on the proposed hybrid model; actual values depend on city, feedstock quality, and plant design

REVENUE FLOW

Feedstock Collection

- Households
- Commercial Establishments
- Markets
- Restaurants & Food Industries
- Industries
- Municipal Corporations (ULBs)

Processing



MRFs & Pre-processing Hybrid Plants (AD,RDF,Pryolysis)

Product & Customers

- CBG → IOCL, BPCL, HPCL, Transport Fleets
- RDF → Cement Plants, Brick Kilns, Steel Industries
- Pyrolysis Oil → Industrial Boilers, Refineries, Marine Fuel Blending
- Organic Fertilizer → Farmers & Agri Companies

Revenue streams

- Fuel Sales (CBG + RDF + Pyrolysis Oil) – 40%
- Municipal Tipping Fees – 20%
- Carbon Credits – 15%
- Organic Fertilizer Sales – 10%
- ESG Reporting Platform – 10%
- Plastic Recovery Certificates – 5%



IMPACT THAT MATTERS

Cleaner Planet. Stronger Economy. Empowered Communities.

ECOVATE transforms waste into value while creating measurable environmental, economic, and social impact at scale.

1. ENVIRONMENTAL IMPACT

Turning Waste Away from Landfills



~1.25 Mn
Tonnes of CO2e Emissions avoided annually*



~3.65 Lakh
Tonnes of waste diverted from landfills every year



~2.10 Lakh
Tonnes of methane emissions prevented annually



~90–110 Mn
Liters of fossil fuel equivalent energy produced annually



~75–80%
Resource recovery through circular waste processing



Cleaner Air, Cleaner Water, Healthier Ecosystem



Every ton of waste processed is a step towards a **sustainable, low-carbon future.**

2. ECONOMIC IMPACT

Creating Value. Driving Growth.



₹60 – 65 Cr
Annual revenue potential (at stabilized operations)



28 – 32%
EBITDA Margin (at stabilized operations)



200+
Direct & Indirect Jobs created per 100 TPD plant



Lower Energy Costs
for industries through affordable, sustainable alternative fuels



Strong ROI
Payback in 4.5 – 5.5 years | IRR: 18 – 22% (Pre-Tax)



Waste becomes a **revenue generator**, fueling industries and the economy.

3. SOCIAL IMPACT

Empowering People. Enriching Communities.



Improved Public Health
Cleaner surroundings lead to reduced air & water pollution and better quality of life.



Livelihood Generation
Formal jobs and skill development across collection, processing & operations.



Stronger Communities
Cleaner cities, reduced littering, and improved urban living conditions.



Capacity Building & Awareness
Promoting waste segregation, circular economy and sustainability mindsets.



Inclusive Growth
Partnerships with ULBs, SHGs, and local entrepreneurs for shared value creation.



Empowered communities build a **cleaner, healthier, and more inclusive India.**

SUPPORTED SDGs

3
GOOD HEALTH AND WELL-BEING

6
CLEAN WATER AND SANITATION

7
AFFORDABLE AND CLEAN ENERGY

8
DECENT WORK AND ECONOMIC GROWTH

11
SUSTAINABLE CITIES AND COMMUNITIES

12
RESPONSIBLE CONSUMPTION AND PRODUCTION

13
CLIMATE ACTION

ALIGNING WITH INDIA'S SUSTAINABILITY GOALS

CUMULATIVE IMPACT (PER 100 TPD PLANT, ANNUAL)



~36,500
Tonnes of waste diverted from landfills



~12,500
Tonnes of CO2e emissions avoided



~2,100
Tonnes of methane emissions prevented



~9 – 11 Mn
Liters of fossil fuel equivalent energy produced



200+
Jobs created directly & indirectly



₹60 – 65 Cr
Annual economic value generated



ECOVATE doesn't just manage waste.

We create a **cleaner environment**, a **stronger economy**, and a **better tomorrow** for generations to come.


Sustainable Today


Profitable Tomorrow


Impactful Forever


Powering a Greener India

CONCLUSION

FROM WASTE CHALLENGE TO LASTING IMPACT

Ecovate's integrated CBG + RDF model is **financially robust, environmentally transformative,** and **socially inclusive** — delivering value for all stakeholders.



KEY TAKEAWAYS



Strong Financials

Attractive returns with IRR ~32%, payback in 2.6 years, and NPV of ₹27.7 Cr (12% WACC).



Environmental Impact

~1.25 Mn tCO2e emissions avoided annually and 3.65 lakh tonnes of waste diverted from landfills.



Social & Community Value

Better public health, higher livelihoods, and inclusive growth for local communities.



Policy Aligned & Future Ready

Fully aligned with national priorities on waste management, clean energy, and climate action.



Scalable & Replicable

A proven hybrid model that can unlock sustainable impact across India.

WHY ACT NOW?

The window for change is open — and the benefits are immediate.



3.65 Lakh

Tonnes of waste diverted annually



~1.25 Mn

tCO2e emissions avoided annually



~90–110 Mn

Liters of fossil fuel equivalent energy produced annually



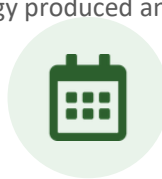
₹26.5 Cr

Year 1 revenue with multiple streams



~ 18 – 22%

Project IRR



4.5-5.5 Years

Simple payback period



Acting now means locking value, driving impact, and staying ahead of the curve.

THE ASK



Partner With Us

Collaborate to implement and scale Ecovate clusters across cities.



Enable & Facilitate

Support through land, approvals, and policy enablers.



Invest in Impact

Join as an investor or strategic partner in a high-impact, high-return opportunity.



Create Shared Value

Together, let's build a cleaner, greener, and more prosperous India.



**YOUR SUPPORT TODAY.
A SUSTAINABLE TOMORROW.**



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